Solutions to 2014 Power Question

1. a. There are $\binom{17}{2} = 136$ possible pairs of dishes, so $\mathcal{F}_{17}$ must have 136 people.

   b. With d dishes there are $\binom{d}{2} = \frac{d^2-d}{2}$ possible pairs, so $n = \frac{d^2-d}{2}$. Then $2n = d^2 - d$, or $d^2 - d - 2n = 0$. Using the quadratic formula yields $d = \frac{1+\sqrt{1+8n}}{2}$ (ignoring the negative value).

   c. The town T' consists of all residents of T who do not know how to make D. Because T is full, every pair of dishes $\{d_i, d_j\}$ in $\text{dish}(T)$ can be made by some resident r_{ij} in T. If $d_i \neq D$ and $d_j \neq D$, then $r_{ij} \in T'$. So every pair of dishes in $\text{dish}(T) \setminus \{D\}$ can be made by some resident of T'. Hence T' is full.

2. a. Paul and Arnold cannot be in the same group, because they both make pie, and Arnold and Kelly cannot be in the same group, because they both make salad. Hence there must be at least two groups. But Paul and Kelly make none of the same dishes, so they can be in the same group. Thus a valid group assignment is

$$\begin{aligned} \text{Paul} &\mapsto 1 \\ \text{Kelly} &\mapsto 1 \\ \text{Arnold} &\mapsto 2. \end{aligned}$$

   Hence $\text{gr}(\text{ARMLton}) = 2$.

   b. Sally and Ross both make calzones, Ross and David both make pancakes, and Sally and David both make steak. So no two of these people can be in the same group, and $\text{gr}(\text{ARMLville}) = 3$.

3. a. Let the dishes be d_1, d_2, d_3, d_4 and let resident r_{ij} make dishes d_i and d_j, where $i < j$. There are six pairs of dishes, which can be divided into nonoverlapping pairs in three ways: $\{1,2\}$ and $\{3,4\}$, $\{1,3\}$ and $\{2,4\}$, and $\{1,4\}$ and $\{2,3\}$. Hence the assignment $r_{12}, r_{34} \mapsto 1$, $r_{13}, r_{24} \mapsto 2$, and $r_{14}, r_{23} \mapsto 3$ is valid, hence $\text{gr}(\mathcal{F}_4) = 3$.

   b. First, $\text{gr}(\mathcal{F}_5) \geq 5$: there are $\binom{5}{2} = 10$ people in $\mathcal{F}_5$, and because each person cooks two different dishes, any valid group of three people would require there to be six different dishes—yet there are only five. So each group can have at most two people. A valid assignment using five groups is shown below.

Residents	Group
r_{12}, r_{35}	1
r_{13}, r_{45}	2
r_{14}, r_{23}	3
r_{15}, r_{24}	4
r_{25}, r_{34}	5

   c. Now there are $\binom{6}{2} = 15$ people, but there are six different dishes, so it is possible (if done carefully) to place three people in a group. Because four people in a single group would require there to be eight different dishes, no group can have more than three people, and

so $15/3 = 5$ groups is minimal. (Alternatively, there are five different residents who can cook dish d_1, and no two of these can be in the same group, so there must be at least five groups.) The assignment below attains that minimum.

Residents	Group
r_{12}, r_{34}, r_{56}	1
r_{13}, r_{25}, r_{46}	2
r_{14}, r_{26}, r_{35}	3
r_{15}, r_{24}, r_{36}	4
r_{16}, r_{23}, r_{45}	5

4. Pick some $n \geq 2$ and a full town $\mathcal{F}_n$ whose residents prepare dishes $d_1, \ldots, d_n$, and let $\text{gr}(\mathcal{F}_n) = k$. Suppose that $f_n : \mathcal{F}_n \to \{1, 2, \ldots, k\}$ is a valid group assignment for $\mathcal{F}_n$. Then remove from $\mathcal{F}_n$ all residents who prepare dish d_n; by problem 1c, this operation yields the full town $\mathcal{F}_{n-1}$. Define $f_{n-1}(r) = f_n(r)$ for each remaining resident r in $\mathcal{F}_n$. If r and s are two (remaining) residents who prepare a common dish, then $f_n(r) \neq f_n(s)$, because f_n was a valid group assignment. Hence $f_{n-1}(r) \neq f_{n-1}(s)$ by construction of f_{n-1}. Therefore f_{n-1} is a valid group assignment on $\mathcal{F}_{n-1}$, and the set of groups to which the residents of $\mathcal{F}_{n-1}$ are assigned is a (not necessarily proper) subset of $\{1, 2, \ldots, k\}$. Thus $\text{gr}(\mathcal{F}_{n-1})$ is at most k, which implies the desired result.

5. Because each chef knows how to prepare exactly two dishes, and no two chefs know how to prepare the same two dishes, each chef is counted exactly twice in the sum $\Sigma\,|\text{chef}_T(D)|$. More formally, consider the set of "resident-dish pairs":

$$S = \{(r, D) \in T \times \text{dish}(T) \mid r \text{ makes } D\}.$$

Count $|S|$ in two different ways. First, every dish D is made by $|\text{chef}_T(D)|$ residents of T, so

$$|S| = \sum_{D \in \text{dish}(T)} |\text{chef}_T(D)|\,.$$

Second, each resident knows how to make exactly two dishes, so

$$|S| = \sum_{r \in T} 2 = 2\,\text{pop}(T).$$

6. Let $D \in \text{dish}(T)$. Suppose that f is a valid group assignment on T. Then for $r, s \in \text{chef}_T(D)$, if $r \neq s$, it follows that $f(r) \neq f(s)$. Hence there must be at least $|\text{chef}_T(D)|$ distinct groups in the range of f, i.e., $\text{gr}(T) \geq |\text{chef}_T(D)|$.

7. For $n = 5$, this result is attained as follows:

Resident	Dishes
Amy	d_1, d_2
Benton	d_2, d_3
Carol	d_3, d_4
Devin	d_4, d_5
Emma	d_5, d_1

For each dish D, note that $\text{chef}_T(D) = 2$. But $\text{gr}(T) > 2$, because if T had at most two groups, at least one of them would contain three people, and choosing any three people will result in a common dish that two of them can cook. Hence T is heterogeneous.

For $n \geq 6$, it suffices to assign dishes to residents so that there are three people who must be in different groups and that no dish is cooked by more than two people, which guarantees that $\text{gr}(T) \geq 3$ and $\text{chef}_T(D) \leq 2$ for each dish D.

Resident	Dishes
Amy	d_1, d_2
Benton	d_1, d_3
Carol	d_2, d_3
Devin	d_4, d_5
Emma	d_5, d_6

Note that Devin's and Emma's dishes are actually irrelevant to the situation, so long as they do not cook any of d_1, d_2, d_3, which already have two chefs each. Thus we can adjust this setup for $n = 7$ by setting Devin's dishes as d_4, d_5 and Emma's dishes as d_6, d_7. (In this last case, Devin and Emma are extremely compatible: they can both be put in a group with anyone else in the town!)

8. a. Because the town is full, each pair of dishes is cooked by exactly one resident, so it is simplest to identify residents by the pairs of dishes they cook. Suppose the first resident cooks (d_1, d_2), the second resident (d_2, d_3), the third resident (d_3, d_4), and so on, until the sixth resident, who cooks (d_6, d_1). Then there are 8 choices for d_1 and 7 choices for d_2. There are only 6 choices for d_3, because $d_3 \neq d_1$ (otherwise two residents would cook the same pair of dishes). For $k > 3$, the requirement that no two intermediate residents cook the same dishes implies that d_{k+1} cannot equal any of $d_1, \ldots, d_{k-1}$, and of course d_k and d_{k+1} must be distinct dishes. Hence there are $8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 = 20{,}160$ six-person resident cycles, not accounting for different starting points in the cycle and the two different directions to go around the cycle. Taking these into account, there are $20{,}160/(6 \cdot 2) = 1{,}680$ distinguishable resident cycles.

   b. Using the logic from 8a, there are $d(d-1)\cdots(d-k+1)$ choices for $d_1, d_2, \ldots, d_k$. To account for indistinguishable cycles, divide by k possible starting points and 2 possible directions, yielding $\frac{d(d-1)\cdots(d-k+1)}{2k}$ or $\frac{d!}{2k(d-k)!}$ distinguishable resident cycles.

9. Note that for every $D \in \text{dish}(T), \text{chef}_T(D) \leq 2$, because otherwise, $r_1, r_2, \ldots, r_n$ could not be a resident cycle. Without loss of generality, assume the cycle is $r_1, r_2, \ldots, r_n$. If n is even, assign resident r_i to group 1 if i is odd, and to group 2 if i is even. This is a valid group assignment, because the only pairs of residents who cook the same dish are (r_i, r_{i+1}) for $i = 1, 2, \ldots, n-1$ and (r_n, r_1). In each case, the residents are assigned to different groups. This proves $\text{gr}(T) = 2$, so T is homogeneous.

   On the other hand, if n is odd, suppose for the sake of contradiction that there are only two groups. Then either r_1 and r_n are in the same group, or for some i, r_i and r_{i+1} are in the

same group. In either case, two residents in the same group share a dish, contradicting the requirement that no members of a group have a common dish. Hence $\text{gr}(T) \geq 3$ when n is odd, making T heterogeneous.

10. a. First note that the condition $|\text{chef}_T(D)| = 2$ for all D implies that $\text{pop}(T) = |\text{dish}(T)|$, using the equation from problem 5. So for the town in question, the population of the town equals the number of dishes in the town. Because no two chefs cook the same pair of dishes, it is impossible for such a town to have exactly two residents, and because each dish is cooked by exactly two chefs, it is impossible for such a town to have only one resident.

The claim is true for towns of three residents satisfying the conditions: such towns must have one resident who cooks dishes d_1 and d_2, one resident who cooks dishes d_2 and d_3, and one resident who cooks dishes d_3 and d_1, and those three residents form a cycle. So proceed by (modified) strong induction: assume that for some $n > 3$ and for all positive integers k such that $3 \leq k < n$, every town T with k residents and $|\text{chef}_T(D)| = 2$ for all $D \in \text{dish}(T)$ can be divided into a finite number of resident cycles such that each resident belongs to exactly one of the cycles. Let T_n be a town of n residents, and arbitrarily pick resident r_1 and dishes d_1 and d_2 cooked by r_1. Then there is exactly one other resident r_2 who also cooks d_2 (because $|\text{chef}_{T_n}(d_2)| = 2$). But r_2 also cooks another dish, d_3, which is cooked by another resident, r_3. Continuing in this fashion, there can be only two outcomes: either the process exhausts all the residents of T_n, or there exists some resident r_m, $m < n$, who cooks the same dishes as r_{m-1} and r_ℓ for $\ell < m - 1$.

In the former case, r_n cooks another dish; but every dish besides d_1 is already cooked by two chefs in T_n, so r_n must also cook d_1, closing the cycle. Because every resident is in this cycle, the statement to be proven is also true for T_n.

In the latter case, the same logic shows that r_m cooks d_1, also closing the cycle, but there are other residents of T_n who have yet to be accounted for. Let $C_1 = \{r_1, \ldots, r_m\}$, and consider the town T' whose residents are $T_n \setminus C_1$. Each of dishes $d_1, \ldots, d_m$ is cooked by two people in C_1, so no chef in T' cooks any of these dishes, and no dish in T' is cooked by any of the people in C_1 (because each person in C_1 already cooks two dishes in the set $\text{dish}(C_1)$). Thus $|\text{chef}_{T'}(D)| = 2$ for each D in $\text{dish}(T')$. It follows that $\text{pop}(T') < \text{pop}(T)$ but $\text{pop}(T') > 0$, so by the inductive hypothesis, the residents of T' can be divided into disjoint resident cycles.
Thus the statement is proved by strong induction.

b. In order for T to be homogeneous, it must be possible to partition the residents into exactly two dining groups. First apply 10a to divide the town into finitely many resident cycles C_i, and assume towards a contradiction that such a group assignment $f : T \to \{1, 2\}$ exists. If $\text{pop}(T)$ is odd, then at least one of the cycles C_i must contain an odd number of residents; without loss of generality, suppose this cycle to be C_1, with residents $r_1, r_2, \ldots, r_{2k+1}$. (By the restrictions noted in part a, $k \geq 1$.) Now because r_i and r_{i+1} cook a dish in common, $f(r_i) \neq f(r_{i+1})$ for all i. Thus if $f(r_1) = 1$, it follows that $f(r_2) = 2$, and that $f(r_3) = 1$, etc. So $f(r_i) = f(r_1)$ if i is odd and $f(r_i) = f(r_2)$ if i is

even; in particular, $f(r_{2k+1}) = f(1)$. But that equation would imply that r_1 and r_{2k+1} cook no dishes in common, which is impossible if they are the first and last residents in a resident cycle. So no such group assignment can exist, and $\mathrm{gr}(T) \geq 3$. Hence T is heterogeneous.

11. a. In problem 5, it was shown that

$$2\,\mathrm{pop}(T) = \sum_{D \in \mathrm{dish}(T)} |\mathrm{chef}_T(D)|\,.$$

Therefore $\sum_{D \in \mathrm{dish}(T)} |\mathrm{chef}_T(D)|$ is even. But if $|\mathrm{chef}_T(D)| = 3$ for all $D \in \mathrm{dish}(T)$, then the sum is simply $3\,|\mathrm{dish}(T)|$, so $|\mathrm{dish}(T)|$ must be even.

b. By problem 6, it must be the case that $\mathrm{gr}(T) \geq 3$. Let $C = \{r_1, r_2, \ldots, r_n\}$ denote a resident cycle such that for every dish $D \in \mathrm{dish}(T)$, there exists a chef in C that can prepare D. Each resident is a chef for two dishes, and every dish can be made by two residents in C (although by three in T). Thus the number of residents in the resident cycle C is equal to $|\mathrm{dish}(T)|$, which was proved to be even in the previous part.

Define a group assignment by setting

$$f(r) = \begin{cases} 1 & \text{if } r \notin C \\ 2 & \text{if } r = r_i \text{ and } i \text{ is even} \\ 3 & \text{if } r = r_i \text{ and } i \text{ is odd.} \end{cases}$$

For any $D \in \mathrm{dish}(T)$, there are exactly three D-chefs, and exactly two of them belong to the resident cycle C. Hence exactly one of the D-chefs r will have $f(r) = 1$. The remaining two D-chefs will be r_i and r_{i+1} for some i, or r_1 and r_n. In either case, the group assignment f will assign one of them to 2 and the other to 3. Thus any two residents who make a common dish will be assigned different groups by f, so f is a valid group assignment, proving that $\mathrm{gr}(T) = 3$.

12. a. From problem 5,

$$2\,\mathrm{pop}(T) = \sum_{D \in \mathrm{dish}(T)} |\mathrm{chef}_T(D)|\,.$$

Because $|\mathrm{chef}_T(D)| = k$ for all $D \in \mathrm{dish}(T)$, the sum is $k \cdot \mathrm{dish}(T)$. Thus $2\,\mathrm{pop}\,T = k \cdot \mathrm{dish}(T)$, and so $k \cdot \mathrm{dish}(T)$ must be even. By assumption, $|\mathrm{dish}(T)|$ is odd, so k must be even.

b. Suppose for the sake of contradiction that there is some n for which the group $R = \{r \in T \mid f(r) = n\}$ has a D-chef for every dish D. Because f is a group assignment and f assigns every resident of R to group n, no two residents of R make the same dish. Thus for every $D \in \mathrm{dish}(T)$, exactly one resident of R is a D-chef; and each D-chef cooks exactly one other dish, which itself is not cooked by anyone else in R. Thus the dishes come in pairs: for each dish D, there is another dish D' cooked by the D-chef in R and no one else in R. However, if the dishes can be paired off, there must be an even number of dishes, contradicting the assumption that $|\mathrm{dish}(T)|$ is odd. Thus for every n, the set $\{r \in T \mid f(r) = n\}$ must be missing a D-chef for some dish D.

c. Let f be a group assignment for T, and let $R = \{r \in T \mid f(r) = 1\}$. From problem 12b, there must be some $D \in \text{dish}(T)$ with no D-chefs in R. Moreover, f cannot assign two D-chefs to the same group, so there must be at least k other groups besides R. Hence there are at least $1 + k$ different groups, so $\text{gr}(T) > k$.

13. a. Fix $D \in \text{dish}(\mathcal{F}_d)$. Then for every other dish $D' \in \text{dish}(\mathcal{F}_d)$, there is exactly one chef who makes both D and D', hence $|\text{chef}_{\mathcal{F}_d}(D)| = d - 1$, which is even because d is odd. Thus for each $D \in \text{dish}(\mathcal{F}_d)$, $|\text{chef}_{\mathcal{F}_d}(D)|$ is even. Because $|\text{dish}(\mathcal{F}_d)| = d$ is odd and $|\text{chef}_{\mathcal{F}_d}(D)| = d - 1$ for every dish in $\mathcal{F}_d$, problem 12c applies, hence $\text{gr}(\mathcal{F}_d) > d - 1$.

Label the dishes $D_1, D_2, \ldots, D_d$, and label the residents $r_{i,j}$ for $1 \le i < j \le d$ so that $r_{i,j}$ is a D_i-chef and a D_j-chef. Define $f : \mathcal{F}_d \to \{0, 1, \ldots, d-1\}$ by letting $f(r_{i,j}) \equiv i+j \bmod d$.

Suppose that $f(r_{i,j}) = f(r_{k,\ell})$, so $i + j \equiv k + \ell \bmod d$. Then $r_{i,j}$ and $r_{k,\ell}$ are assigned to the same group, which is a problem if they are different residents but are chefs for the same dish. This overlap occurs if and only if one of i and j is equal to one of k and ℓ. If $i = k$, then $j \equiv \ell \bmod d$. As j and ℓ are both between 1 and d, the only way they could be congruent modulo d is if they were in fact equal. That is, $r_{i,j}$ is the same resident as $r_{k,\ell}$. The other three cases ($i = \ell$, $j = k$, and $j = \ell$) are analogous. Thus f is a valid group assignment, proving that $\text{gr}(\mathcal{F}_d) \le d$. Therefore $\text{gr}(\mathcal{F}_d) = d$.

b. In problem 4, it was shown that the sequence $\text{gr}(\mathcal{F}_2), \text{gr}(\mathcal{F}_3), \ldots$ is nondecreasing. If d is even, $\text{gr}(\mathcal{F}_d) \ge \text{gr}(\mathcal{F}_{d-1})$, and because $d-1$ is odd, problem 13a applies: $\text{gr}(\mathcal{F}_{d-1}) = d-1$. Hence $\text{gr}(\mathcal{F}_d) \ge d - 1$. Now it suffices to show that $\text{gr}(\mathcal{F}_d) \le d - 1$ by exhibiting a valid group assignment $f : \mathcal{F}_d \to \{1, 2, \ldots, d - 1\}$.

Label the dishes $D_1, \ldots, D_d$, and label the residents $r_{i,j}$ for $1 \le i < j \le d$ so that $r_{i,j}$ is a D_i-chef and a D_j-chef. Let $R = \{r_{i,j} \mid i, j \ne d\}$. That is, R is the set of residents who are not D_d-chefs. Using 1c, R is a full town with $d-1$ dishes, so from 12a, it has a group assignment $f : R \to \{1, 2, \ldots, d-1\}$. For each $D_i \in \text{dish}(\mathcal{F}_d)$, $i \ne d$, $|\text{chef}_R(D_i)| = d-2$. Because there are $d - 1$ groups and $|\text{chef}_R(D_i)| = d - 2$, exactly one group n_i must not contain a D_i-chef for each dish D_i.

It cannot be the case that $n_i = n_j$ for $i \ne j$. Indeed, suppose for the sake of contradiction that $n_i = n_j$. Without loss of generality, assume that $n_i = n_j = 1$ (by perhaps relabeling the dishes). Then any resident $r \in R$ assigned to group 1 (that is, $f(r) = 1$) would be neither a D_i-chef nor a D_j-chef. The residents in R who are assigned to group 1 must all be chefs for the remaining $d - 3$ dishes. Because each resident cooks two dishes, and no two residents of group 1 can make a common dish,

$$|\{r \in R \mid f(r) = 1\}| \le \frac{d-3}{2}.$$

For each of the other groups $2, 3, \ldots, d - 1$, the number of residents of R in that group is no more than $(d - 1)/2$, because there are $d - 1$ dishes in R, each resident cooks two dishes, and no two residents in the same group can make a common dish. However,

because $d-1$ is odd, the size of any group is actually no more than $(d-2)/2$. Therefore

$$\begin{aligned}
|R| &= \sum_{k=1}^{d-1} |\{r \in R \mid f(r) = k\}| \\
&= |\{r \in R \mid f(r) = 1\}| + \sum_{k=2}^{d-1} |\{r \in R \mid f(r) = k\}| \\
&\leq \frac{d-3}{2} + \sum_{k=2}^{d-1} \frac{d-2}{2} \\
&= \frac{d-3}{2} + \frac{(d-2)^2}{2} \\
&= \frac{d^2-3d+1}{2} < \frac{d^2-3d+2}{2} = |R| \,.
\end{aligned}$$

This is a contradiction, so it must be that $n_i \neq n_j$ for all $i \neq j$, making f a valid group assignment on $\mathcal{F}_d$. Hence $\mathrm{gr}(\mathcal{F}_d) = d-1$.